Ananta Narayan Shrestha

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Summary

Software engineer with 6+ years of experience, including 4 years shipping FDA-regulated software at Philips Ultrasound R&D — where I was the sole developer of a real-time audio module for the Lumify iOS app (Swift, C++, AVFoundation/AVAudioEngine), a clinical ultrasound product on the App Store — owning it from requirements through delivery. Deep systems background in C++/Rust (real-time, concurrency, performance), suited to iOS roles in media, audio/video, or performance-critical native code.

Experience

Software Engineer — Dec 2025 – Present
Exito Tech Inc. — New York, USA

- Building an AI-powered chatbot on AWS using Anthropic Claude and retrieval-augmented generation (RAG) to improve response grounding and quality.

Software Engineer I & II — Jun 2018 – Jun 2022
Philips India Ltd. (Ultrasound R&D – Lumify iOS App) — Bangalore, India

- Sole developer of a multi-threaded, low-latency (5 ms) real-time audio module for the Lumify iOS app: owned requirements gathering with clinical scientists, framework selection (AVFoundation/AVAudioEngine), and delivery of the shipped feature in an FDA-regulated App Store medical product.
- Engineered real-time audio-buffer processing end to end — tuning buffer sizes and sample rates against latency constraints — integrating a cross-platform C++ core with native Swift layers.
- Streamlined UI development in Swift by replacing manual XIBs with programmatic StackViews, cutting new feature development time by 80% while shipping 10+ new UI controls across 10+ device sizes and orientations.
- Implemented real-time diagnostic algorithms (Quad AFI, High Q Doppler) in C++ under 5 ms latency constraints, supporting the launch of a major new product line; mentored new recruits across Agile projects.

Research Software Engineer — Aug 2023 – Sep 2025
National Renewable Energy Laboratory — Colorado, USA

- Contributed to RouteE Compass (Rust routing engine enabling Google Maps' eco-routing): TOML configuration injection, core crate refactor, and an SI-unit energy model; recognized with the U.S. DOE-VTO AMC Team Award 2025.
- Built a large-scale (~1 TB) HPC data-processing pipeline for EV infrastructure grid simulation, cutting execution time from 10 hours to 30 minutes (95%) using Polars and Parquet.

Software Engineer Intern II — May 2023 – Aug 2023
Yahoo Inc. — Illinois, USA

- Integrated four fuzzing engines (Google FuzzTest, LibFuzzer, AFL++, HonggFuzz) into Yahoo's production DNS infrastructure in C++17, identifying and reporting two critical memory-management vulnerabilities.

Projects & Open Source

Energy Policy RAG — github.com/iantei/energy-rags

- Retrieval-Augmented Generation system: Medallion ETL on Polars/Parquet, ChromaDB, LangChain, Docker/Ollama GPU inference.

Open Source: NatLabRockies/routee-compass (merged PR #364, gzip query-file support); nanvix research OS (signal handling, POSIX fixes).

Education

Master of Computer Science, GPA 4.0/4.0 — Arizona State University, Tempe, AZ — May 2024

B.Tech. Computer Science & Engineering — MNNIT, Prayagraj, India — May 2018

Technical Skills

iOS & Mobile:	Swift, UIKit, AVFoundation / AVAudioEngine, MVVM, Xcode, cross-platform C++ integration, real-time audio
Languages:	Swift, C++, Python, Rust, C#, SQL
Cloud & DevOps:	AWS (EC2, S3, SQS, Lambda), Docker, GitHub Actions, CI/CD, CMake, Linux
AI/LLM:	LangChain, RAG architecture, Anthropic API; daily use of Claude Code, Cursor, Copilot
Practices:	Automated testing, technical documentation, mentoring & onboarding, Agile, SDLC

Awards & Publications

- U.S. DOE–VTO AMC Team Award 2025 — RouteE & FASTSim enabled Google Maps' eco-routing feature.
- Co-author: Powell, B., Shrestha, A., et al., "Where are all the Electric Vans?", ASCE ICTD 2025.
- Philips Ultrasound R&D Hackathon: Winner 2019–20; Runner-up 2018–19; Team Up to Win 2022.